

Learning Whom to Test: Per-Case Credit Assignment for Laboratory Test Allocation in Arbovirus Surveillance

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Abstract

When dengue and chikungunya co-circulate, clinical triage cannot reliably tell them apart, and laboratory confirmation is too scarce to be applied to every suspected case. Deciding *which* suspects to test is a sequential decision made in epidemic time: tests take days to return, results change what should be done next, and hundreds of patients are open at once. We cast this problem as a Markov decision process in a simulator grounded in surveillance data—SEIR dynamics with literature parameters, spatial case distributions estimated by kriging from notifications in Rio de Janeiro and Recife, noisy clinicians, delayed laboratory results, and suspects that have neither disease. Two findings frame the problem. First, in real data the location of a case barely reveals its disease (a Bayes classifier using location alone is right 52–61% of the time across two cities and eleven epidemic years), so tests are the main source of information. Second, standard deep reinforcement learning fails to learn to test: because decisions about many patients interleave, the value of a test is diluted across dozens of unrelated steps (the correlation between the standard return and the test’s outcome is -0.02 , against 0.99 for a return computed along the patient’s own timeline). We propose a per-case variant of proximal policy optimization that computes advantages along each patient’s timeline, discounted in epidemic days, without changing the environment or the reward. With identical observations, per-case credit raises the mean episode reward from $+684$ to $+2692$ in a two-disease variant of the simulator (four training seeds; $P(\text{better}) = 1.00$ in every paired outbreak). With non-arboviral suspects, where a negative test no longer settles a case, the learned policy matches the best expert rule we could write—sequential testing guided by the clinician’s suspicion—in reward ($+2387$ vs. $+2330$; difference $+57$, 95% CI $[-128, +221]$) and accuracy (0.938 vs. 0.943), while ordering 7% fewer tests in every evaluated outbreak; the standard agent collapses to inaction. Transferred without retraining to a second city (Recife, 2016 and 2021 epidemics), the agent stays tied with the rule in reward with 10–11% fewer tests, at a cost of 1–2 accuracy points. Results in the full environment are preliminary (one training seed per arm).

Keywords: reinforcement learning; credit assignment; arboviruses; diagnostic testing; resource allocation; epidemiological surveillance.

1 Introduction

Arboviral diseases are among the most persistent public-health burdens in tropical countries such as Brazil [1]. Dengue, chikungunya and Zika share the *Aedes aegypti* vector and cause large cyclical epidemics that periodically overwhelm health services, with costs that range from lost productivity to hospital and emergency care [2, 3]. When these viruses co-circulate, their clinical pictures overlap, which hampers real-time surveillance [4].

The case definition used by the Brazilian Ministry of Health—sudden fever with arthralgia, myalgia or rash—is sensitive enough to capture most infections, but has low specificity for the

viral etiology [5]. Distinguishing dengue from chikungunya from history and physical examination alone is error-prone [6], and a sizeable fraction of suspects turns out to have neither disease. Laboratory confirmation resolves the ambiguity, but during fast-growing outbreaks testing every suspect is financially and logistically infeasible, so managers must ration tests. In practice this rationing is often reactive: tests are spent on clinically obvious cases while atypical or epidemiologically strategic ones remain unconfirmed [4].

Allocating laboratory tests is therefore a *sequential decision problem under uncertainty*. Testing a suspect, relying on epidemiological evidence, waiting for more information, or assigning a final diagnosis consumes finite resources and shapes the quality of future surveillance [7, 8]. Because the true state of the epidemic is only partially observed—through reporting delays, clinical noise and asymptomatic infection—good policies must reason from incomplete information and update decisions over time [8, 9]. Reinforcement learning (RL) is a natural framework for such policies, and simulation environments with a standard interface make it possible to train and compare them [10, 11].

We deliberately keep the decisions inside *epidemic time*: suspects arrive day by day, tests return after a laboratory delay, and a case comes back to the decision-maker when its result is available. This choice is what makes the problem realistic, and it is also what makes it hard to learn. With hundreds of cases open at once, the consequence of a decision about one patient arrives interleaved with the consequences of dozens of decisions about others. Standard RL estimators chain consecutive steps of the trajectory, so the value of ordering a test must travel through many unrelated decisions before it is credited—and in our setting it does not survive the trip.

Contributions.

1. **A data-grounded simulator of test allocation in epidemic time.** Dengue and chikungunya outbreaks follow SEIR dynamics with literature parameters; case locations are drawn from kriging surfaces estimated from notifications in Rio de Janeiro and Recife; suspects include non-arboviral febrile illness; clinicians are noisy; laboratory results return after five days and send the case back to the agent (Section 2).
2. **Evidence that location is weakly informative in real data.** Across two cities and eleven epidemic years, the best classifier that sees only the location of a case is right 52–61% of the time (52–57% in every year with more than a few hundred cases of each disease), far from the 94% of idealized spatial generators. The test, not the map, is the main source of information (Section 5.1).
3. **A diagnosis of why standard deep RL fails, and a fix.** We measure that the standard return is uncorrelated with the outcome of a test (-0.02), while a return computed along the patient’s own timeline is almost perfectly correlated (0.99). A per-case variant of PPO that estimates advantages along each patient’s timeline, discounted in epidemic days, makes the agent investigate without changing the environment or the reward (Sections 3.3 and 5.2).
4. **An honest evaluation.** We compare learned policies against strong fixed rules—including sequential testing guided by the clinician—with paired outbreaks and a hierarchical, paired bootstrap. The learned policy matches the best rule with fewer tests; we show that which rule is best depends on the case mix (Sections 4 and 5.3).

2 Simulation environment

The environment implements the Gymnasium interface [10, 12]. Each episode simulates 300 days of concurrent dengue and chikungunya outbreaks in one city. Every parameter reported below

is set in version-controlled configuration files, and the environment’s behavior is locked by an automated test suite.

2.1 Epidemic dynamics

Each disease follows a frequency-dependent SEIR model in a closed population [13, 14]:

$$\dot{S} = -\beta \frac{SI}{N}, \quad \dot{E} = \beta \frac{SI}{N} - \sigma E, \quad \dot{I} = \sigma E - \gamma I, \quad \dot{R} = \gamma I, \quad (1)$$

with $\sigma = 1/\text{latency}$, $\gamma = 1/\text{infectious period}$ and $\beta = R_0\gamma$. Notifications follow the flow σE (onset of the infectious phase, which we take as symptom onset); the cumulative curve is integrated numerically and daily new cases are its rounded increments.

Because both viruses pass through the mosquito, the interval between successive cases includes the extrinsic incubation. Rather than a full host–vector model, we use an *effective* human latent period that absorbs both incubations, so that the model’s generation time (latency plus infectious period) matches the serial interval observed in the field (Table 1). For dengue, the intrinsic incubation averages 5.9 days [15], human infectiousness lasts about five days after symptom onset [16], and space–time clustering is strongest at 15–17 days [17]; R_0 in Rio de Janeiro was estimated at 1.70 (2002) and 1.25 (2012) [18]. For chikungunya in Rio, $R_0 = 1.56$ (95% CI 1.46–1.67) assuming a 14-day generation time [19]; only the sum of latency and infectious period is anchored by data. Each episode draws R_0 uniformly within these ranges, with a population of 300 per disease and 1% initially infected. We validate the generator against properties that can be checked against theory—the growth rate implied by R_0 and the final-size equation $z = 1 - e^{-R_0 z}$ (Figure 1)—rather than against the shape of the curves, which can look plausible when the epidemiology is wrong.

Disease	Effective latency	Infectious period	Generation time	R_0
Dengue	11 d	5 d	16 d	1.25–1.70
Chikungunya	8 d	6 d	14 d	1.46–1.67

Table 1: SEIR parameters. R_0 is drawn uniformly within the range at each episode. Sources: [15–19].

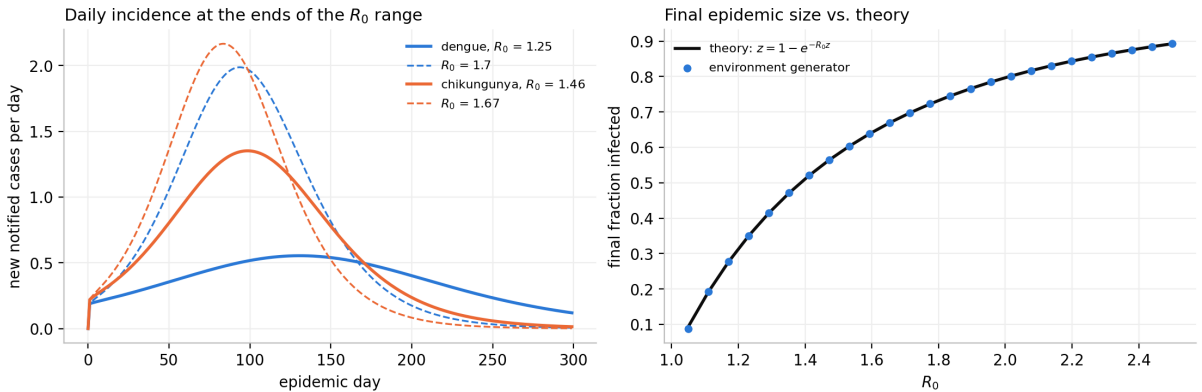


Figure 1: Epidemic dynamics. Left: daily incidence at the ends of the R_0 range for each disease; epidemics last several months. Right: final fraction infected produced by the generator against the final-size equation.

2.2 Where cases occur: spatial surfaces from surveillance data

Cases are placed on a 400×400 grid by an interchangeable spatial generator. The default generator uses surfaces estimated by ordinary kriging from georeferenced notifications: counts are aggregated in 500 m cells, the log-counts are interpolated, and the result is normalized into a spatial probability per disease, $P(\text{cell} \mid \text{disease})$, from which each day’s cases are sampled. At every episode one of four rigid transformations (identity, 180° rotation, horizontal or vertical mirror) is applied *jointly* to both surfaces, so the agent cannot memorize one city’s geography while the relative geometry of the two diseases—and hence the difficulty—is preserved.

Rio de Janeiro. The reference surfaces use dengue and chikungunya notifications of Rio de Janeiro in 2015–2016 with residence coordinates. Chikungunya barely circulated in Rio in 2015 (70 notifications, against about 14 thousand in 2016), so alternative Rio scenarios combine dengue from 2015 or 2016 with chikungunya from 2016.

Recife. As a second city we use the open notification microdata of Recife [20], which cover dengue (2013–2025) and chikungunya (2015–2025). Each record carries the final classification (including discarded cases), the confirmation criterion, laboratory results and the residence street and neighborhood, but no coordinates. We geocode each record against the city’s official neighborhood polygons and street segments [21]: the street name is matched (after normalizing abbreviations) to the streets that cross the reported neighborhood, or a neighboring one, and a point is drawn along the matched street; when the street is not found, a point is drawn uniformly within the neighborhood. Of 144,175 confirmed or discarded notifications of Recife residents, 72% were placed on the reported street within the reported neighborhood, 6% on the same street in an adjacent neighborhood, 22% at neighborhood level, and fewer than 1% could not be placed. We build one surface per year (2015–2025) from cases confirmed by any criterion, and set the probability to zero outside the municipality, which also removes the sea. Chikungunya notifications of 2015–2016 coded with the legacy dengue classification scale (4,888 records without any chikungunya-specific result) are excluded as ambiguous.

How informative is location? What matters for the decision problem is not whether each disease has sharp foci, but whether the two diseases occupy *different* places. We summarize this with the accuracy of the Bayes classifier that sees only the location of a case, with equal priors: $\frac{1}{2} \sum_{\text{cells}} \max\{P(\text{cell} \mid \text{dengue}), P(\text{cell} \mid \text{chik})\}$. It equals 0.5 when location carries no information and 1 when it determines the disease. In Rio 2015–2016 it is 0.544: each surface has clear foci, but the foci coincide (correlation 0.85 between the two surfaces; in 90% of the cells one disease is at most 1.4 times more likely than the other), because both follow where people live and where the vector is (Figure 2, top). The idealized generator used in earlier work—two truncated Gaussian foci—reaches 0.937.

Controlling spatial informativeness. To study sensitivity to this quantity we sharpen the surfaces with a temperature τ ($p \propto p^\tau$; $\tau = 0$ is uniform), which moves the Bayes accuracy from 0.50 ($\tau = 0$) through 0.544 ($\tau = 1$) to 0.94 ($\tau = 32$). Sharpening reaches high separability only by concentrating each disease: half of the dengue cases fall in 34% of the cells at $\tau = 1$, in 1% at $\tau = 8$ and in 8 of 13,195 cells at $\tau = 32$ (Figure 2, bottom). Intermediate values preserve the shape of the foci; $\tau \geq 16$ is an artificial limit rather than a plausible epidemic.

2.3 Case mix: suspects that are neither disease

In real triage, many suspects have another febrile illness. The simulator adds non-arboviral suspects at 25% of the notified cases, uniform in space (they do not follow the vector’s foci) and proportional in time to the arboviral cases of each day. This proportion is consistent with the

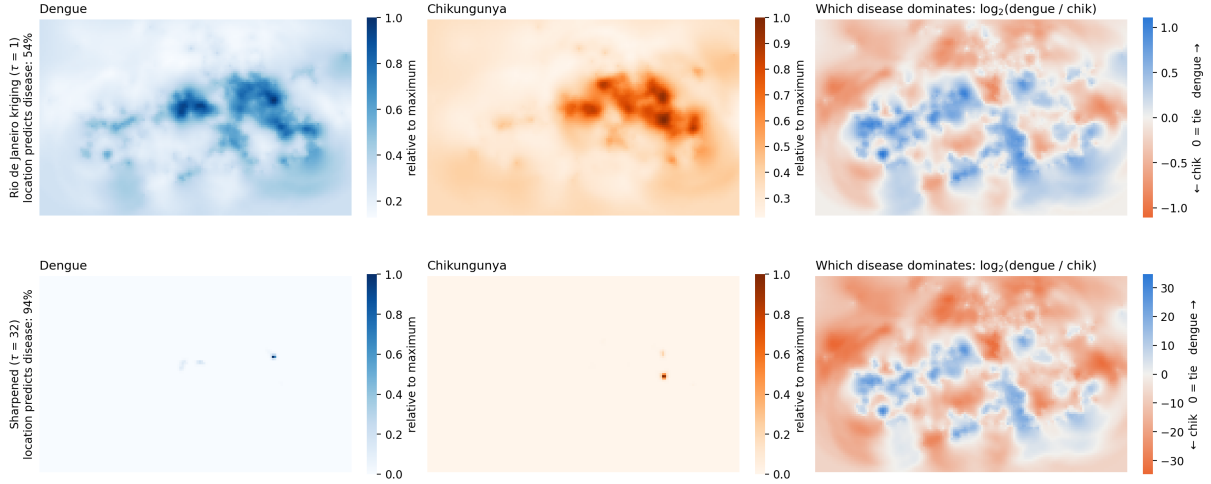


Figure 2: Spatial surfaces of dengue and chikungunya, each on its own scale, and the log-ratio between them. Top: kriging on Rio de Janeiro notifications (2015–2016); each disease has sharp foci, but the ratio stays between $0.5\times$ and $2\times$ and location predicts the disease 54% of the time. Bottom: sharpened surfaces ($\tau = 32$); location becomes highly informative only because each disease collapses onto a handful of cells.

Recife data: in the epidemic years (2015, 2016 and 2021), discarded notifications are 18–30% of the total. These cases give depth to the investigation: a negative dengue test no longer implies chikungunya, a second test becomes informative, and concluding “not an arbovirus” becomes a legitimate decision.

Two-disease variant. An earlier version of our kriging generator omitted the non-arboviral suspects, so every suspect had one of the two diseases. We keep this configuration as an ablation of the case mix—the *two-disease variant*—because it isolates how the case mix changes which policy is optimal (Section 5.3). Several robustness analyses were run in this variant and are reported as such.

2.4 Clinicians

The true disease is hidden from the agent and replaced by a noisy clinical label. Each episode draws a clinician whose accuracy is uniform in $[0.50, 0.95]$ and applies to both diseases. Non-arboviral suspects are recognized as such at triage with probability 0.15; otherwise the clinician labels them dengue or chikungunya with equal probability—if they were easy to recognize, they would not be in the queue of suspects. The environment also supports Beta-distributed sensitivity and specificity with controllable mean and dispersion. The clinician turns out to be the dominant source of variance between episodes (Section 4).

2.5 Laboratory tests and epidemiological confirmation

Laboratory tests follow RT-PCR as the diagnostic gold standard: sensitivity 0.95, specificity 0.98 and 2% indeterminate results, which leave the working diagnosis unchanged. **[TODO: anchor the RT-PCR parameters in references; they vary with the kit, the day of collection and the serotype.]** Results become available five days after collection. Epidemiological confirmation simulates a field investigation: a case is confirmed if, within a radius of 20 grid cells, there is more than one other case already confirmed *by tests the agent ordered*. It therefore uses only information the surveillance system has, and it makes each positive test an investment in

future confirmations nearby. Under a policy that tests every case, it confirms 19.3% of cases with 80.1% precision, against 57.1% accuracy of the clinical label.

2.6 Decision process

The epidemic advances in days; within each day, the cases awaiting a decision are presented to the agent one at a time, so each agent step is a decision about one case (Figure 3). The agent observes a six-channel spatial tensor summarizing notified cases, tests and confirmed cases; the coordinates of the current case; and a 16-dimensional context vector with the clinical label, the results of tests already done, epidemiological confirmation, the local density of confirmed cases, and a two-number summary of the evidence accumulated about the clinician (agreement between test results and clinical labels so far, and how much evidence supports it). This summary tracks the clinician’s true accuracy with correlation 0.99; estimating it requires spending some tests early, which creates a natural exploration–exploitation trade-off.

Actions. For each case the agent chooses among seven actions: **no action**, which costs nothing and lets the clinician’s diagnosis stand; **investigate**, by ordering a dengue test, a chikungunya test, or epidemiological confirmation; and **conclude**, by asserting dengue, chikungunya or “other”. The asserted class comes from the action itself, so the agent can disagree with the clinician without ordering a test.

Case lifecycle. When the agent investigates, the sample is collected immediately but the result matures five days later; **when the result arrives, the case returns to the agent**, who decides with the result in hand. A case leaves the queue when it is concluded, when the agent takes no action on it, or when the episode ends; it can return at most twice, after which only concluding actions are allowed. Without returns, ordering a test would mean giving up the decision about that case, and testing and deciding would be mutually exclusive.

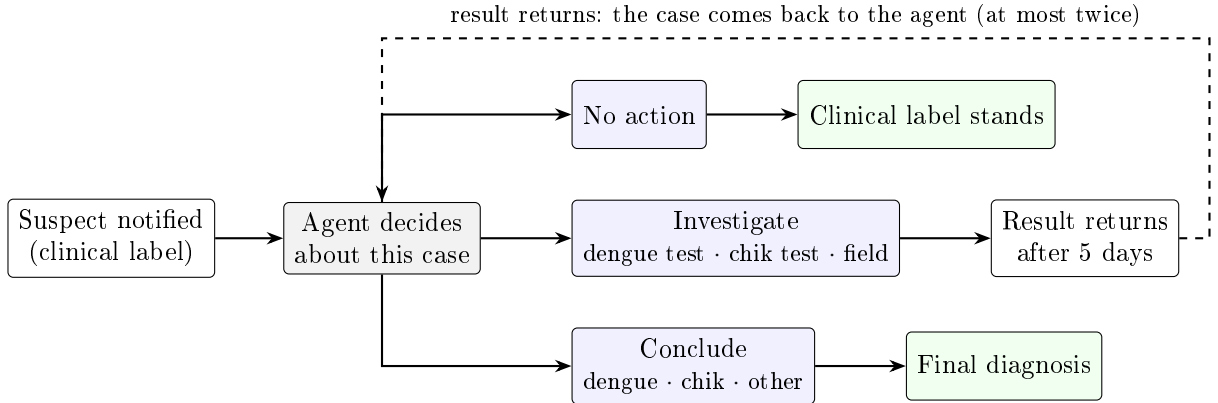


Figure 3: Lifecycle of a suspected case. Each agent step is a decision about one case; hundreds of cases are open at the same time, so the steps of one case are interleaved with those of many others.

2.7 Reward

The reward has three layers. **Immediate cost:** each test or field investigation costs 4.0; no action and concluding cost nothing. **Decision outcome:** a concluding action earns +10 if correct, −30 if it discards a real arbovirus case (a surveillance false negative), and −20 for other errors; concluding an investigated case earns a +2 bonus. **Terminal score:** at the end of the episode each case contributes +1 if its final diagnosis is correct and −3 otherwise, whether or not

it was tested, so a test is rewarded for the information it brings and not for being ordered. A case whose investigation was opened and never concluded costs -10 , charged when it is abandoned; taking no action on an uninvestigated case is not abandonment but acceptance of the clinical diagnosis, judged by the terminal score.

The test cost governs whether an allocation problem exists: with cheap tests, testing everyone is optimal; with expensive ones, the best policy is to do nothing. At 4.0, investigating pays with a narrow margin—the regime in which choosing *which* cases to investigate matters most. **[TODO: repeat the test-cost sweep in the full environment; the current calibration was done with the idealized generator.]**

Per-case decomposition. At each step, the environment also reports which case the step’s reward belongs to, on which epidemic day the decision was made, and whether the case is finished; each case’s terminal share is attributed to it exactly once. The per-case shares sum exactly to the episode reward, so the metric used to compare agents is unchanged. This decomposition is the raw material for the credit assignment of Section 3.3.

3 Agents

3.1 Fixed policies

Fixed policies establish what can be achieved without learning; without them, a trained agent can look competent merely by beating chance.

- **Clinical only:** never acts; the clinician’s label stands.
- **Confirm-all:** concludes every case with the clinical label, without testing.
- **Test-once:** orders one dengue test per case and concludes with the result.
- **Test-twice:** orders both tests for every case and concludes with both results.
- **Sequential, dengue first:** orders a dengue test; orders the chikungunya test only if the first is not positive; concludes.
- **Sequential, clinician-guided:** as above, but first tests for the disease the clinician suspects (dengue if the clinician suspects neither).

The sequential rules are the natural expert policies between test-once and test-twice: a positive result settles the case, a negative or indeterminate one calls for the other test.

3.2 Learning agents

Deep Q-network. We first trained a DQN [22] with a convolutional encoder. It is kept as a control: once stabilized with a Huber loss, it converges reliably to a degenerate policy of inaction (Section 5.2).

Proximal policy optimization. PPO [23] optimizes a stochastic policy π_θ with the clipped objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t [\min(\rho_t(\theta)\hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)], \quad \rho_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, \quad (2)$$

where the advantage $\hat{A}_t$ is usually obtained by generalized advantage estimation (GAE) [24] over *consecutive* steps of the trajectory:

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \quad \hat{A}_t = \sum_{k \geq 0} (\gamma \lambda)^k \delta_{t+k}. \quad (3)$$

3.3 Per-case credit assignment

The problem. Consecutive steps of the trajectory belong to different patients. Between a test and the decision it informs there are about 30 decisions about *other* cases; the test itself only returns -4 , and its benefit appears in a later step that belongs to the same case but is surrounded by unrelated rewards. Equation (3) forces that value through all of them, accumulating their noise. We measured the effect directly: at the step of each test, the correlation between the return and whether the test led to a correct diagnosis is -0.019 for the standard return discounted per step, and 0.994 for the return of the case itself discounted per epidemic day. The signal is in the trajectory; the standard definition of the return erases it.

This also explains why three earlier attempts failed. Potential-based reward shaping [25] does not apply: the next state is already another patient, so the potential cancels within the step, and with hundreds of open investigations the drift $(1 - \gamma)\Phi$ exceeds the intended increment for any scale. Retroactive credit would require rewriting rewards already delivered. Making tests cheaper doubled the margin of investigating and did not make the agent order a single test. The obvious alternative—one episode per case—would remove the interleaving by destroying the epidemic dynamics, which is the object of this work.

The method. Each transition carries the episode and case identifiers, the epidemic day d of the decision and the case’s share r^{case} of the reward. *Per-case PPO* groups the transitions of a batch by (episode, case) and computes GAE along each patient’s timeline, with discounting raised to the number of *days* between decisions rather than steps. For the successive decisions j of one case:

$$\begin{aligned} \text{disc}_j &= \gamma^{d_{j+1}-d_j}, \\ \delta_j &= r_j^{\text{case}} + \text{disc}_j V(s_{j+1}) - V(s_j), \\ \hat{A}_j &= \delta_j + \text{disc}_j \lambda \hat{A}_{j+1}. \end{aligned} \tag{4}$$

At the last step of a case in the batch there is no future if the case is finished; if the batch cut its timeline, we bootstrap from its own value with one day of discount. The structure mirrors credit assignment in multi-agent systems, with each case as an agent sharing the world with the others. **[TODO: cite multi-agent credit-assignment literature.]** Two invariances are verified by automated tests: the per-case shares reconstruct the episode reward, so the evaluation metric is unchanged; and the advantage of a case does not depend on the interleaving—inserting 30 noisy decisions about other cases between two steps of case X leaves X ’s advantage unchanged.

Limitation. Per-case credit ignores the externality of a test: a positive result improves epidemiological confirmation for *future* cases nearby, and that benefit does not flow back to the case that paid for the test. The agents barely use epidemiological confirmation, so the effect is currently small.

3.4 Architecture and training

All deep agents share the same network: convolutional layers process the spatial tensor and end in adaptive pooling to $64 \times 6 \times 6$, and a dense branch processes the case coordinates and context vector [26]. Since the pooling discards resolution, the map is delivered at 100×100 , which reduced observations sixteen-fold at no measurable cost. We use Tianshou [27] with 300,000 steps per run, learning rate 3×10^{-4} , $\gamma = 0.99$, $\lambda = 0.95$, $\epsilon = 0.2$, entropy coefficient 0.02, batches of 256 and eight parallel training environments. The action mask that forces a conclusion after two returns is applied to the actor’s logits—otherwise PPO silently ignores it. Rewards are scaled by 0.10 during training only.

To separate the effect of credit assignment from that of temporal information, we trained three arms with the same network and hyperparameters (Table 2): $A \times C$ isolates credit assignment with identical observations; $B \times C$ isolates four optional temporal features (fraction of the horizon elapsed, cases notified today, 7-day trend, age of the case), all computed from what surveillance observes.

Arm	Advantage	Context	Training seeds
A	standard GAE (Eq. 3)	16	4 (two-disease); 1 (full)
B	per case, in days (Eq. 4)	16 + 4 temporal	3 (two-disease)
C	per case, in days (Eq. 4)	16 (as A)	4 (two-disease); 1 (full)

Table 2: PPO arms.

4 Evaluation protocol

Paired outbreaks. Agents are evaluated on 10 test outbreaks, each identified by a fixed seed that reproduces the same epidemic, the same clinician and the same clinical noise for every agent, so comparisons are *paired*. This matters: the clinician drawn in an episode explains most of the variance—its accuracy correlates at 0.97 with the episode reward of both the clinical policy and a trained agent. A second set of 12 never-used seeds tests generalization. We always evaluate the *final* checkpoint; selecting the best checkpoint during training selects noise (it scored 1,000–1,460 points above the final checkpoint for DQN).

Hierarchical, paired bootstrap. There are two sources of variance: the seed that trained the agent and the outbreak it is evaluated on. Confidence intervals come from a hierarchical bootstrap [28] that resamples training seeds and then outbreaks; outbreak resampling is *shared* across arms within each replicate, preserving the pairing, while training seeds—independent runs—are resampled separately per arm. Following Agarwal et al. [29], we report the mean, the interquartile mean, the paired difference between arms and the probability of improvement $P(X > Y)$, the chance that an episode of X beats the same episode of Y over all pairs of training seeds (ties count $1/2$), with 95% percentile intervals from 10,000 replicates. With three or four training seeds the upper level has little resolution, so intervals are honest about outbreaks and optimistic about training seeds; with one seed (full environment) they cover outbreaks only.

Metrics. The primary metric is the episode reward; secondary metrics are three-class diagnostic accuracy (dengue, chikungunya, other) and the number of laboratory tests per episode. In the two-disease variant no case is “other”, so accuracy is effectively binary.

5 Results

5.1 In real data, location barely reveals the disease

Recife offers eleven years of both diseases in one city, with epidemics of very different composition—dengue-dominated in 2015–2016 and 2019, chikungunya-dominated in 2021 (Figure 4, left). In every year, the Bayes accuracy of location alone stays close to Rio’s 0.544: between 0.520 and 0.573 from 2016 to 2025 (Figure 4, right). The highest value, 0.605 in 2015, comes from only 201 confirmed chikungunya cases, which makes that surface artificially concentrated. The places where one disease dominates change from year to year (Figure 5), but the magnitude of the difference stays small. Recife 2016, the largest epidemic with both diseases, is strikingly similar

to Rio 2015–2016 (Bayes accuracy 0.557 vs. 0.544; correlation between the two diseases’ surfaces 0.81 vs. 0.85).

These estimates are stable to the random placement of geocoded points: across five geocoding seeds, the Bayes accuracy of each year varies by at most ± 0.01 . The correlation between the two surfaces is noisier in years with few cases (standard deviation up to 0.16) and is not interpreted there.

The implication for the decision problem is direct. The idealized generator, with location 94% informative, lets an agent diagnose from the map; real surveillance data do not. In earlier work with that generator, a PPO agent reached a positive reward without ordering a single test, by disagreeing with the clinician based on location. In real geography, the laboratory test is the main source of information, and the allocation problem is real.

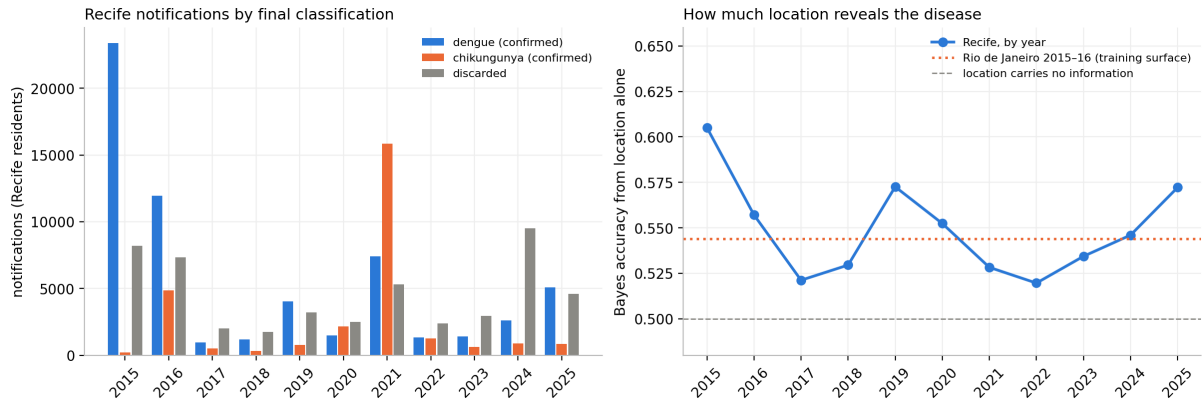


Figure 4: Recife, 2015–2025. Left: notifications of residents by final classification. Right: accuracy of a Bayes classifier that sees only the location of a case, per year, against the Rio de Janeiro surface used for training (dotted) and the no-information level (dashed).

5.2 Standard deep RL does not learn to investigate; per-case credit does

DQN and standard PPO. DQN was unstable, with rewards oscillating by up to 6,300 points between neighboring evaluations. A Huber loss stabilized it unambiguously (from ± 1340 to ± 9 across seeds) and made it *worse*: stabilized, it converged reliably to 66.7% inaction and no tests (-4164 ± 9). With the same environment, network and action mask, PPO with standard GAE was the first learned agent to beat doing nothing ($+632 \pm 43$), with monotonic training curves. Learning rate, buffer size, n -step returns and target-update frequency did not move DQN measurably. PPO, however, hardly investigated: 7 to 37 tests per episode in the Rio geography, against about 370 for test-once.

Per-case credit. Table 3 and Figure 6 show the three arms in the two-disease variant. Three conclusions follow.

The gain comes from credit assignment. With identical observations, C outperforms A by +2009 [$+1197, +2882$] with $P(C > A) = 1.00$: in every outbreak, for every pair of training seeds, the per-case agent wins. It moves from 25 to 284 tests per episode and gains 20 accuracy points. A also collapses in specific outbreaks (down to -2200 in those with poor clinicians), where it inherits the clinician’s errors because it barely tests; C never collapses.

The temporal features do not contribute. B and C are indistinguishable (-43 [$-196, +85$], $P = 0.47$). This does not show that temporal information is useless, only that these four features added nothing on top of per-case credit.

The per-case agent matches the best fixed rule with fewer tests. Without non-arboviral suspects, a single test settles a case and test-once is the best fixed policy. C’s reward is statistically

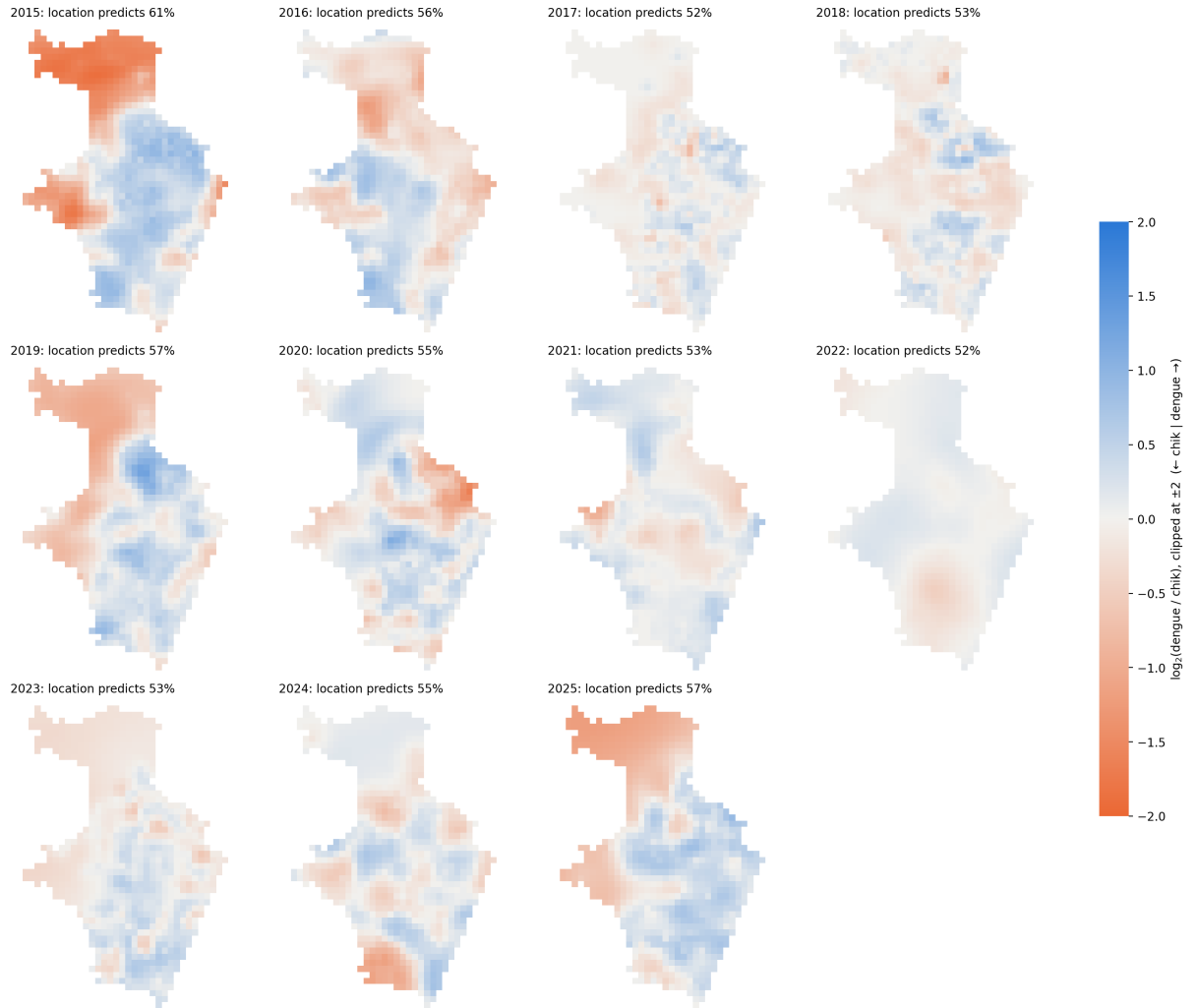


Figure 5: Where each disease dominates in Recife, year by year: $\log_2(P(\text{cell} | \text{dengue})/P(\text{cell} | \text{chik}))$ from kriging surfaces, clipped at ± 2 . The map changes between years, but the contrast remains weak.

indistinguishable from it ($-90 [-250, +67]$) with 84 fewer tests per episode (-23%) and 2.2 fewer accuracy points. On 12 unseen outbreaks the pattern holds: $C \times A = +1653 [+1072, +2313]$, $P = 1.00$; $C \times \text{test-once} = +6 [-124, +149]$. An input ablation without retraining shows that zeroing the entire spatial map changes only 0–2% of the agent’s decisions: the policy decides from the case context, consistent with the weak spatial signal of Section 5.1.

Policy	Seeds	Reward [95% CI]	Accuracy [95% CI]	Tests [95% CI]
Test-once (best fixed rule)	–	+2782 [+2668, +2900]	0.960 [0.953, 0.966]	369 [353, 384]
C: per-case credit	4	+2692 [+2523, +2842]	0.937 [0.923, 0.950]	284 [223, 335]
B: per-case credit + time	3	+2649 [+2413, +2842]	0.931 [0.908, 0.946]	266 [206, 321]
Test-twice	–	+1051 [+969, +1135]	0.950 [0.945, 0.954]	737 [706, 768]
A: standard GAE	4	+684 [−146, +1498]	0.739 [0.674, 0.803]	25 [5, 64]
Confirm-all	–	+346 [−656, +1335]	0.710 [0.633, 0.789]	0
Clinical only	–	−64 [−184, +54]	0.710 [0.633, 0.789]	0

Table 3: Two-disease variant (Rio de Janeiro surfaces), 10 paired outbreaks; final checkpoints; hierarchical, paired bootstrap. Paired contrasts: $C \times A = +2009 [+1197, +2882]$, $P = 1.00$; $B \times C = -43 [-196, +85]$, $P = 0.47$; $C \times \text{test-once} = -90 [-250, +67]$, $P = 0.38$.

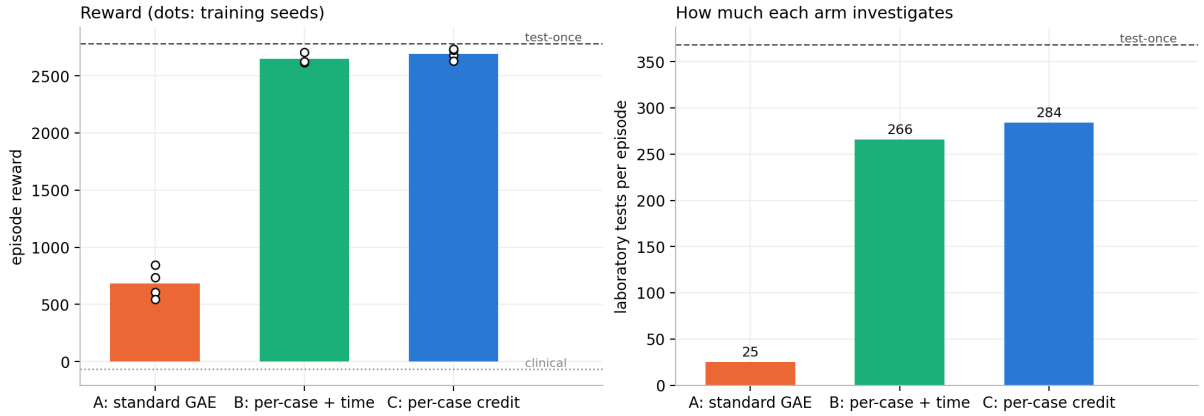


Figure 6: The three PPO arms in the two-disease variant. Left: mean episode reward (dots are training seeds). Right: laboratory tests per episode. Dashed lines: test-once; dotted: clinical only.

5.3 With non-arboviral suspects: the case mix decides the best rule

Adding non-arboviral suspects changes the problem, not just the numbers (Table 4). With only two diseases, a negative dengue test implies chikungunya and test-once is optimal. With a quarter of the suspects being neither, a negative test is ambiguous: test-once drops from +2782 to +322 and 0.756 accuracy, and investigation must go deeper. The idealized generator, which always had non-arboviral suspects, shows the same ordering as the full environment—the reordering is caused by the case mix, not by geography. The best fixed rule is now sequential testing guided by the clinician (+2330): testing first for the suspected disease settles most cases with one test when the clinician is right, and saves 249 tests per episode over test-twice (+1105 [+1010, +1190] in reward).

Arms A and C were retrained in the full environment with the same procedure, one training seed each (Table 5, Figure 7).

The per-case agent matches the best expert rule with fewer tests. C reaches +2387 against +2330 for the clinician-guided sequential rule: the difference, +57 [−128, +221], is not

Fixed policy	Kriging, two-disease	Kriging, full	Idealized generator
Test-once	+2782	+322	+473
Test-twice	+1051	+1225	+1249
Sequential, clinician-guided	–	+2330	[TODO: run]
Confirm-all	+346	–1734	–1821
Clinical only	–64	–344	–355

Table 4: Fixed policies with SEIR dynamics, 10 paired outbreaks. Removing non-arboviral suspects (two-disease variant) is what makes a single test sufficient; the idealized generator, which has them, orders the policies as the full environment does.

distinguishable from zero ($P(C > \text{rule}) = 0.70$), and neither is the difference in accuracy (-0.005 $[-0.016, +0.004]$). C orders 54 fewer tests per episode $[-62, -46]$, and fewer in every one of the 10 outbreaks. Against the other fixed rules C is clearly better: +413 $[+189, +635]$ over sequential dengue-first, +1163 $[+980, +1330]$ over test-twice with 303 fewer tests, and +2065 over test-once with 18 more accuracy points. The agent was never told about test order or about the clinician’s reliability; it learned a policy as good as the best rule we could design, and a slightly more frugal one.

The standard agent collapses. With standard GAE, A orders about six tests per episode, its accuracy falls below the clinician’s, and its reward falls below doing nothing (-1627 $[-2436, -800]$ against clinical only). The gap between the arms grows from +2009 in the two-disease variant to +4359 $[+3434, +5276]$. A plausible explanation, not yet tested, is that the value of investigating now depends on a longer chain—test, result, second test, result, conclusion—which is exactly the kind of chain that interleaving erases from the standard estimator.

Policy	Seeds	Reward [95% CI]	Accuracy [95% CI]	Tests [95% CI]
C: per-case credit	1	+2387 $[+2186, +2564]$	0.938 $[0.929, 0.947]$	675 $[633, 716]$
Sequential, clinician-guided	–	+2330 $[+2231, +2414]$	0.943 $[0.936, 0.950]$	729 $[683, 774]$
Sequential, dengue first	–	+1974 $[+1768, +2182]$	0.940 $[0.932, 0.948]$	804 $[779, 828]$
Test-twice	–	+1225 $[+1131, +1315]$	0.938 $[0.933, 0.943]$	978 $[938, 1014]$
Test-once	–	+322 $[+219, +431]$	0.756 $[0.750, 0.763]$	489 $[469, 507]$
Clinical only	–	–344 $[-459, -225]$	0.576 $[0.521, 0.635]$	0
Confirm-all	–	–1734 $[-2694, -730]$	0.576 $[0.521, 0.635]$	0
A: standard GAE	1	–1972 $[-2897, -1028]$	0.563 $[0.511, 0.619]$	6 $[2, 11]$

Table 5: Full environment (Rio de Janeiro surfaces, with non-arboviral suspects), 10 paired outbreaks. With one training seed per arm, intervals cover outbreak variation only. Paired contrasts: $C \times \text{clinician-guided sequential} = +57$ $[-128, +221]$, $P = 0.70$, tests -54 $[-62, -46]$; $C \times A = +4359$ $[+3434, +5276]$, $P = 1.00$.

5.4 Robustness

Other years and spatial informativeness. Without retraining, agents trained on the Rio 2015–2016 surfaces were evaluated on surfaces from other years (dengue 2015 with chikungunya 2016, and Rio 2016 alone) and on sharpened surfaces. These analyses were run in the two-disease variant. Across years the results are practically unchanged (C: +2692, +2710, +2721; $C \times A$ between +2009 and +2077, $P \geq 0.98$), as expected from surfaces that are all about 54% informative. Across the whole range of spatial informativeness, from 0.50 to 0.94, the advantage of per-case credit holds ($C \times A$ between +2000 and +2350, P between 0.89 and 1.00; Figure 8). C’s reward stays flat: an agent trained where location is uninformative does not exploit it when

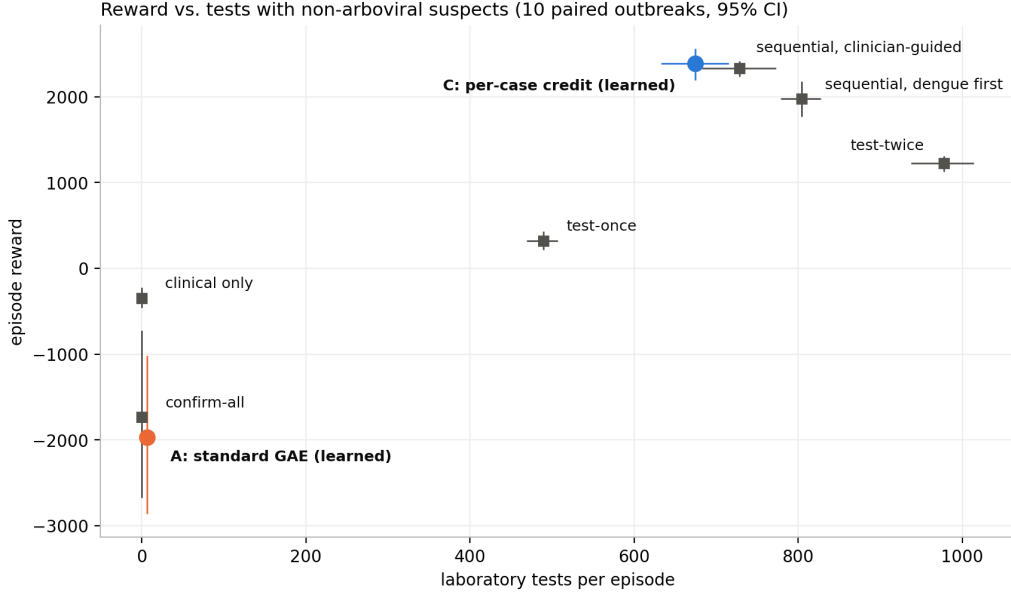


Figure 7: Reward against tests per episode in the full environment (means with 95% intervals over 10 paired outbreaks). The per-case agent sits next to the best expert rule, with fewer tests; the standard agent collapses near inaction.

it becomes informative. At $\tau = 8$, two of C’s four seeds degrade, a sign of fragility outside the training distribution.

A second city. The full-environment agents, trained on Rio de Janeiro surfaces, were evaluated without retraining on the Recife surfaces of 2016 (the largest dengue-dominated epidemic with both diseases) and 2021 (chikungunya-dominated), on the same 10 paired outbreaks (Table 6). Fixed policies do not use location, so their results are identical across cities. The pattern of the training city holds in both years: the per-case agent is statistically tied with the clinician-guided sequential rule in reward, orders 10–11% fewer tests in every outbreak, and outperforms the standard agent by about +4000 with $P = 1.00$. The cost of transfer shows up in accuracy: C falls 1.2–1.7 points below the rule, a small but significant gap that was not significant in the training city (−0.5 points). One caveat applies to any comparison between cities: the environment grid is 400×400 for every city, so the 20-cell radius of epidemiological confirmation corresponds to about 1 km in Recife and 3.5 km in Rio de Janeiro.

Surface	C reward	C $\times$ rule [95% CI]	Δ tests [95% CI]	Δ accuracy [95% CI]	C $\times$ A
Rio 2015–16 (training)	+2387	+57 [−128, +221]	−54 [−62, −46]	−0.005 [−0.016, +0.004]	+4359
Recife 2016	+2235	−95 [−278, +88]	−76 [−88, −63]	−0.017 [−0.027, −0.006]	+3778
Recife 2021	+2345	+15 [−185, +205]	−81 [−95, −67]	−0.012 [−0.022, −0.001]	+4223

Table 6: Zero-shot transfer to a second city (full environment, one training seed per arm, 10 paired outbreaks). “Rule” is the clinician-guided sequential policy (+2330, 729 tests, 0.943 accuracy in every city, since fixed policies ignore location). $P(C > A) = 1.00$ in all rows.

6 Discussion

The obstacle to learning a testing policy was not where it first seemed. For much of this project, “the agent does not test” was treated as a problem of exploration, reward scale or algorithm. Part of it was: a stabilized DQN reliably learns inaction, and PPO was the first learned agent to beat

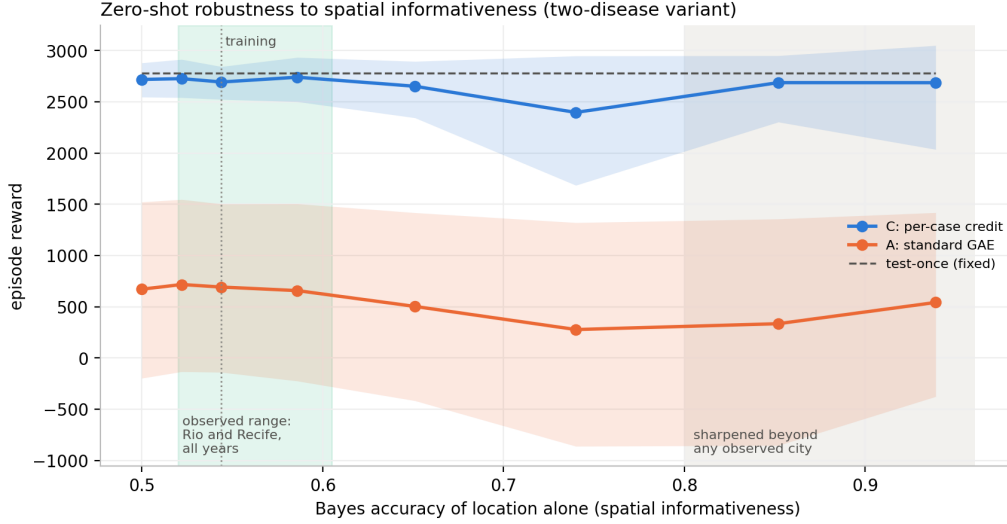


Figure 8: Zero-shot evaluation as spatial informativeness varies (two-disease variant; 95% intervals over training seeds and outbreaks). The green band is the range observed in Rio de Janeiro and Recife; the gray band requires sharpening beyond any observed city.

doing nothing. But the decisive bottleneck was credit assignment. When decisions about many patients interleave in epidemic time, the standard return carries almost no information about whether a test was useful; the return along the patient’s own timeline carries almost all of it. Estimating advantages per case, in days, recovers that signal without touching the environment or the reward.

What the agent learns depends on the problem it faces. In the two-disease variant, where one test always suffices, it learns *whether* to investigate and matches test-once with 23% fewer tests. With non-arboviral suspects, investigation has depth, and the agent learns a policy as good as sequential testing guided by the clinician—the best rule we could write—while ordering 7% fewer tests. Matching an expert rule is a meaningful result for a policy learned from reward alone, but it is not a claim of superiority; the headroom that remains is large (in an earlier configuration, an oracle that knew the true disease reached 0.977 accuracy with about a third of the tests). A useful by-product of the analysis is itself practical: among fixed rules, testing first for the disease the clinician suspects is substantially better than always testing dengue first (+356 in reward and 75 fewer tests per episode).

The weak spatial signal in real surveillance data explains why the agents ignore the map (zeroing it changes 0–2% of decisions) and cautions against conclusions drawn from idealized geographies. It also points to where geography might matter: in the *changing* location of dominance from year to year (Figure 5) and in epidemiological confirmation from confirmed neighbors, which the agents currently barely use.

6.1 Limitations

- **Training seeds.** Results in the full environment have one training seed per arm; their intervals do not capture variation between training runs. In the two-disease variant, the standard deviation between seeds was about 50 points for C and 130 for A, small relative to the differences reported, but more seeds are needed.
- **Robustness in the full environment.** The analyses across years, spatial informativeness and unseen outbreaks were run in the two-disease variant and must be repeated in the full environment.

- **Simulated clinicians and tests.** Clinician accuracy is symmetric between diseases and drawn per episode; test parameters are fixed; non-arboviral suspects are uniform in space. Recife’s discarded notifications could replace the uniform assumption with an empirical surface, but a notification discarded for dengue may be chikungunya, so they are not a pure “other” class.
- **Data.** Rio surfaces are built from notified cases and Recife surfaces from confirmed cases; Recife points are geocoded to street or neighborhood level. In 2015–2016 Recife was the epicenter of the Zika epidemic, and part of the dengue confirmed by clinical–epidemiological criteria in those years is likely Zika.
- **Externality of tests.** Per-case credit does not credit a test for its effect on future epidemiological confirmations nearby.
- **Capacity and seasonality.** Scarcity is modeled through test cost, not a daily laboratory capacity; the SEIR model has no seasonal forcing.

6.2 Future work

The immediate steps are more training seeds in the full environment, the robustness analyses repeated there, and a per-case analysis of which cases the agent tests twice—which would turn “the agent chooses which cases need a second test” into a measured statement. Beyond that: training on sharpened or multi-city surfaces to test whether per-case credit also wins when geography is informative and whether the agent learns to use it; additional years of Rio de Janeiro notifications; synthetic populations from the 2022 Brazilian census [30] to build scenarios for other cities with realistic population density; a daily laboratory capacity, which would turn scarcity into explicit prioritization; and a credit assignment that accounts for the externality of positive tests.

7 Conclusion

We framed the allocation of laboratory tests in arbovirus surveillance as a sequential decision in epidemic time, in a simulator grounded in surveillance data from two Brazilian cities. In real data, the location of a case barely reveals whether it is dengue or chikungunya, so tests are the main source of information—and standard deep RL fails to learn to use them, because the value of a test is lost among interleaved decisions about other patients. Estimating advantages along each patient’s timeline, in epidemic days, makes the agent investigate: it matches the best fixed rule in each setting with fewer tests, where the standard agent collapses toward inaction. Which rule is best, in turn, depends on the case mix: when a quarter of suspects have neither disease, a single test no longer suffices, and testing first for the disease the clinician suspects becomes the policy to beat.

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A Work in progress (internal; remove before submission)

- [TODO: Train ≥ 4 seeds per arm in the full environment and redo the hierarchical bootstrap.]
- [TODO: Repeat in the full environment: unseen outbreaks, other years, spatial-informativeness sweep, clinician quality, test cost, laboratory delay.]
- [TODO: Per-case analysis of which cases C tests twice (as a function of the first result, the clinical label and the evidence about the clinician).]
- [TODO: Fixed sequential rule on the idealized generator (Table 4); Recife evaluation with more training seeds and more years; training on Recife surfaces.]
- [TODO: Train A and C on sharpened surfaces ($\tau = 4, 8$) or on a mixture of cities; design a sharpening that amplifies the ratio between diseases while preserving each map's shape.]
- [TODO: Related work: RL for testing and epidemic control; multi-agent credit assignment; test allocation in surveillance.]
- [TODO: Anchor RT-PCR parameters in references; redo the test-cost sweep in the full environment; derive the information ceiling with three classes and two tests.]
- [TODO: Check flagged bibliography entries (marked in refs.bib).]
- [TODO: Compare the proposed flow with the municipal health department's decision flow.]